

Dynamic

Video-133

Programming



Note :- This playlist is only for explanation of Dns & solutions.
See my "DP Concepts & Dns"
Playlist for understanding
DP from scratch...



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CSwithMIK



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Motivation :-

Memoize the lessons from your past failures so you never have to process the same mistakes twice.

The optimal path to success is built simply by mastering one subproblem at a time.



MIK...

3129. Find All Possible Stable Binary Arrays I

Medium Topics Companies Hint

You are given 3 positive integers zero, one, and limit.

A binary array arr is called stable if:

- The number of occurrences of 0 in arr is exactly zero.
- The number of occurrences of 1 in arr is exactly one.
- Each subarray of arr with a size greater than limit must contain both 0 and 1.

Return the *total* number of **stable** binary arrays.

Since the answer may be very large, return it **modulo** $10^9 + 7$.

Example 1:

Input: zero = 1, one = 1, limit = 2

Output: 2

Explanation:

The two possible stable binary arrays are [1,0] and [0,1], as both arrays have a single 0 and a single 1, and no subarray has a length greater than 2.

Example 2:

Input: zero = 1, one = 2, limit = 1

Output: 1

Explanation:

The only possible stable binary array is [1,0,1].

Note that the binary arrays [1,1,0] and [0,1,1] have subarrays of length 2 with identical elements, hence, they are not stable.

Example 3:

Input: zero = 3, one = 3, limit = 2

Output: 14

Explanation:

All the possible stable binary arrays are [0,0,1,0,1,1], [0,0,1,1,0,1], [0,1,0,0,1,1], [0,1,0,1,0,1], [0,1,0,1,1,0], [0,1,1,0,0,1], [0,1,1,0,1,0], [1,0,0,1,0,1], [1,0,0,1,1,0], [1,0,1,0,0,1], [1,0,1,0,1,0], [1,0,1,1,0,0], [1,1,0,0,1,0], and [1,1,0,1,0,0].

Constraints :-

$1 \leq \text{zero}, \text{one}, \text{limit} \leq 200$

Thought Process

zero $\rightarrow$ exactly count of 0s ✓✓
 one $\rightarrow$ exactly count of 1s ✓✓

Each subarray having size $>$ limit must contain both 0 and 1

$$\text{limit} = 2$$

1 000 111 01100

3, 4, 5

$$\text{limit} = 2$$

1 0001

ee You cannot have more than Limit consecutive
0s/1s "

$$\text{Zero} = 3$$

$$\text{one} = 2$$

$$\text{limit} = 2$$

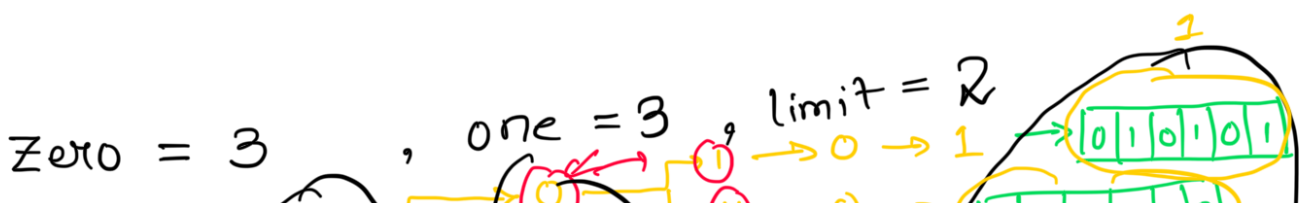
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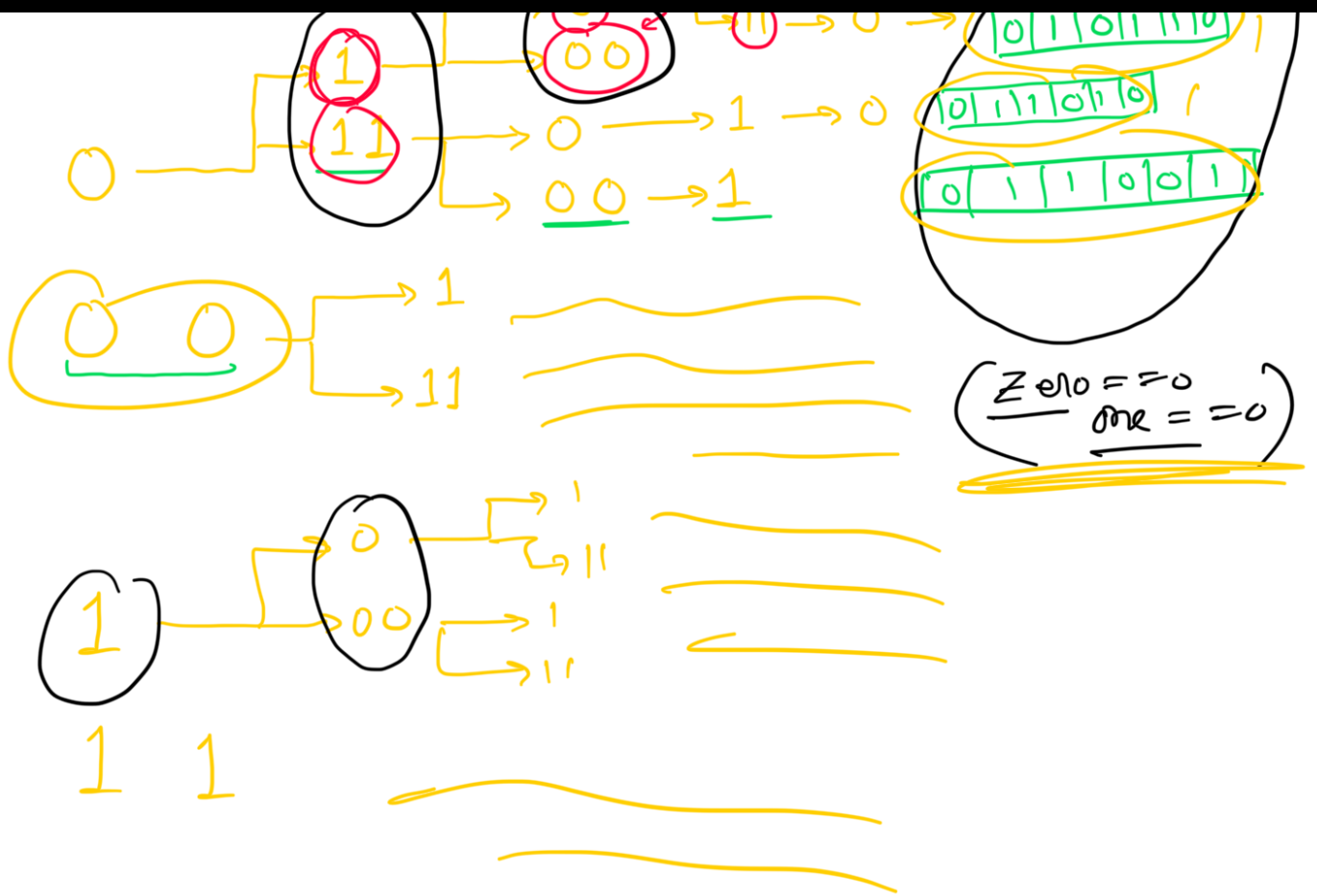
3
4X
5X



Did I explain the problem
statement to long ???

That was the only hard part
of the problem. Now, see how easy
it will be 😊





Recursion.

Zero = 3

limit = 2

min(Zero, limit)

min (one, limit)

Story To Code :-

Solve (onesLeft, zerosLeft, bool lastWasOne, limit)

if (onesLeft == 0 && zerosLeft == 0)
return 1;

// explore 0s
⇒ if (lastWasOne == True) {
 for (int len = 1; len <= min(limit, zerosLeft); len++)
 {
 result += Solve(onesLeft,
 zerosLeft - len, False, limit);
 }
}

else {
 // Explore 1s
 for (int len = 1; len <= min(limit, onesLeft); len++) {
 result += Solve(onesLeft - len, zerosLeft,
 True, limit);
 }


```

    }
    return result;
}

```

StartWithOne = Solve (one, zero, False, limit);
 StartWithZero = Solve (one, zero, True, limit);

Bottom UP

(Using exact same code of Recursion).

```

int solve(int onesLeft, int zerosLeft, bool lastWasOne, int limit) {
    if(onesLeft == 0 && zerosLeft == 0) {
        return 1;
    }

    if(t[onesLeft][zerosLeft][lastWasOne] != -1) {
        return t[onesLeft][zerosLeft][lastWasOne];
    }

    int result = 0;
    // p = 1 / p = 0

```

↓ ↓ ↓
 t[][][]

t[0][0][1] = 1


```

if(lastWasOne == true) { //explore 0s
    for(int len = 1; len <= min(zerosLeft, limit); len++) {
        result = (result + solve(onesLeft, zerosLeft - len, false, limit)) % M;
    }
} else { //explore 1s
    for(int len = 1; len <= min(onesLeft, limit); len++) {
        result = (result + solve(onesLeft - len, zerosLeft, true, limit)) % M;
    }
}

return t[onesLeft][zerosLeft][lastWasOne] = result;
}

```

$f(0)[0][0] = 1$

f
 $\left[\begin{array}{c} \downarrow \\ \text{onesLeft} \end{array} \right]$
 $\left[\begin{array}{c} \downarrow \\ \text{zerosLeft} \end{array} \right]$
 $\left[\begin{array}{c} 0/1 \\ \downarrow \\ \text{lastWasOne} \end{array} \right]$

→ for (onesLeft = 0; onesLeft <= One; onesLeft++) {

→ for (zerosLeft = 0; zerosLeft <= Zero; zerosLeft++) {

if (onesLeft == 0 && zerosLeft == 0) {
continue;
}

int result = 0; //exp 0s

for (len = 1; len <= min(zerosLeft, limit); len++) {

result = (res + f [onesLeft][zerosLeft - len][0]) % M;

}

f [onesLeft][zerosLeft][1] = result;

lastWasOne = true;

result = 0; //exp 1s

```

lastWaOne = fact -
    0
    {
        for (len = 1; len <= min(onesLeft, limit); len++) {
            result = (res + f[onesLeft-len][zeroLeft][1]) % M;
        }
        f[onesLeft][zeroLeft][0] = result;
    }
}

```

$a = f[one][zero][0];$

$b = f[one][zero][1];$

$\text{return } (a+b) \% M;$

the 1990s, the number of people in the world who are undernourished has increased from 250 million to 800 million (FAO 1996).

There are a number of reasons why the world's population is becoming increasingly undernourished. One of the main reasons is the rapid growth of the world's population. The world's population is expected to reach 8 billion by the year 2025, up from 5 billion in 1987 (UN 1994).

Another reason is the increasing demand for food. As the world's population grows, the demand for food increases. This is particularly true in developing countries, where the population is growing rapidly and the demand for food is increasing.

A third reason is the increasing demand for meat. As the world's population grows, the demand for meat increases. This is particularly true in developed countries, where the population is growing rapidly and the demand for meat is increasing.

A fourth reason is the increasing demand for fish. As the world's population grows, the demand for fish increases. This is particularly true in developed countries, where the population is growing rapidly and the demand for fish is increasing.

A fifth reason is the increasing demand for dairy products. As the world's population grows, the demand for dairy products increases. This is particularly true in developed countries, where the population is growing rapidly and the demand for dairy products is increasing.

A sixth reason is the increasing demand for processed food. As the world's population grows, the demand for processed food increases. This is particularly true in developed countries, where the population is growing rapidly and the demand for processed food is increasing.

A seventh reason is the increasing demand for alcohol. As the world's population grows, the demand for alcohol increases. This is particularly true in developed countries, where the population is growing rapidly and the demand for alcohol is increasing.

A eighth reason is the increasing demand for tobacco. As the world's population grows, the demand for tobacco increases. This is particularly true in developed countries, where the population is growing rapidly and the demand for tobacco is increasing.

A ninth reason is the increasing demand for drugs. As the world's population grows, the demand for drugs increases. This is particularly true in developed countries, where the population is growing rapidly and the demand for drugs is increasing.

A tenth reason is the increasing demand for pornography. As the world's population grows, the demand for pornography increases. This is particularly true in developed countries, where the population is growing rapidly and the demand for pornography is increasing.

A eleventh reason is the increasing demand for gambling. As the world's population grows, the demand for gambling increases. This is particularly true in developed countries, where the population is growing rapidly and the demand for gambling is increasing.

A twelfth reason is the increasing demand for prostitution. As the world's population grows, the demand for prostitution increases. This is particularly true in developed countries, where the population is growing rapidly and the demand for prostitution is increasing.

the 1990s, the number of people in the world who are under 15 years of age has increased by 1.2 billion (United Nations 1999).

There is a growing awareness of the need to address the needs of children in the 21st century. The United Nations Convention on the Rights of the Child (1989) has been signed by 112 countries, and the United Nations Millennium Declaration (2000) has set out a commitment to 'ensure that all children, everywhere, have access to primary education by the year 2015'. The United Nations Secretary-General Kofi Annan (1999) has called for 'a new global compact for children' to ensure that the rights of children are protected and promoted.

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